Modelling Moon Phases · Tier Pack · Aspire

Learning goal: I can test a 3D model of the Moon's orbit to record all eight phases and evaluate the model.

Aspire Route (Book First)

Calculate the approximate number of days between each of the 8 model positions in a 28-day cycle, and evaluate two geometric limitations of the ball-and-torch model.

Help: 28 days divided by 8 positions is __ days per step. Two limitations are that the torch is smaller than the __ and the distances are not to __.

Optional Make It · Aspire

Make a Model Redesign Brief

Explain how the model would change if an observer viewed the Moon from the Southern Hemisphere, and calculate approximately how many days pass between each of the 8 positions (28 / 8 = 3.5 days).

Word Help

waxing crescent: the phase after New Moon when a thin sliver of light grows on the right

waxing gibbous: the phase before Full Moon when more than half of the face is lit and growing

waning gibbous: the phase after Full Moon when more than half is lit and shrinking

waning crescent: the final thin sliver of light before

the cycle returns to New Moon

Exit Check

In our model, what did the torch, the observer, and the ball represent?

Write the four main phases in order starting from New Moon.

State one strength and one limitation of the ball-and-torch model.